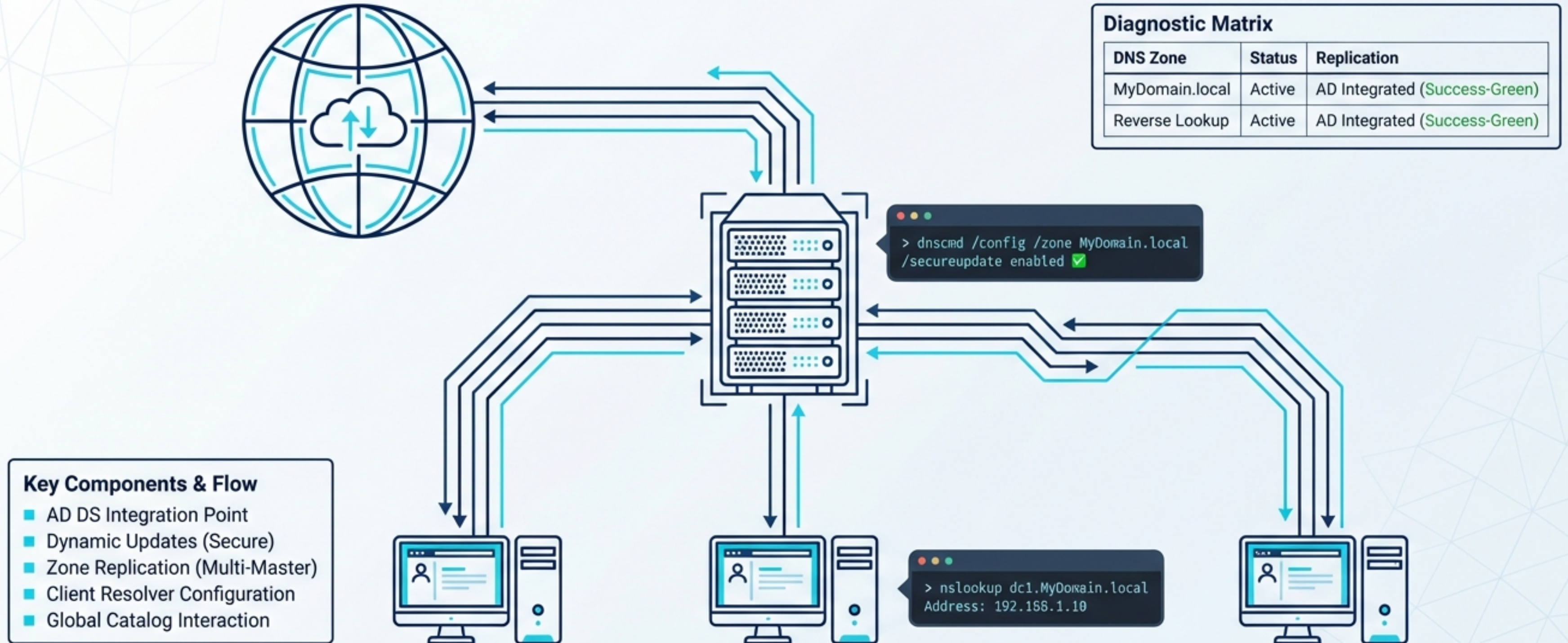
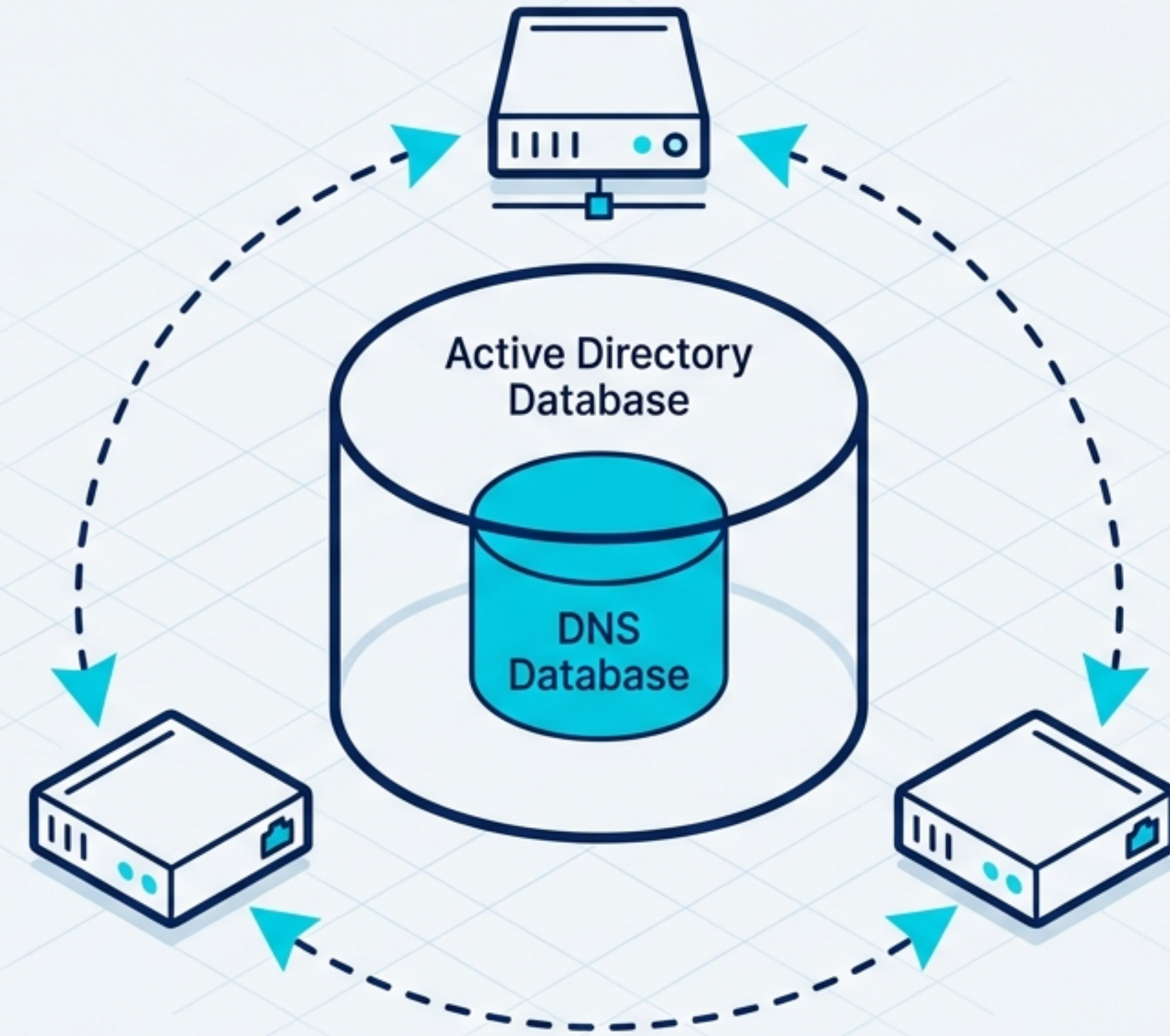


Configuring Active Directory Integrated DNS

Standard Operating Procedure & Architectural Reference



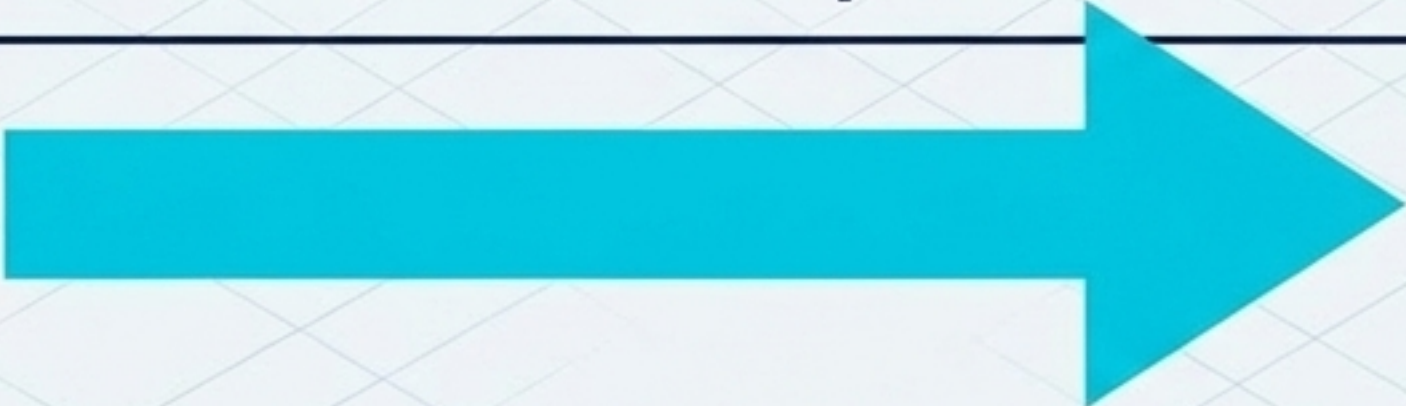
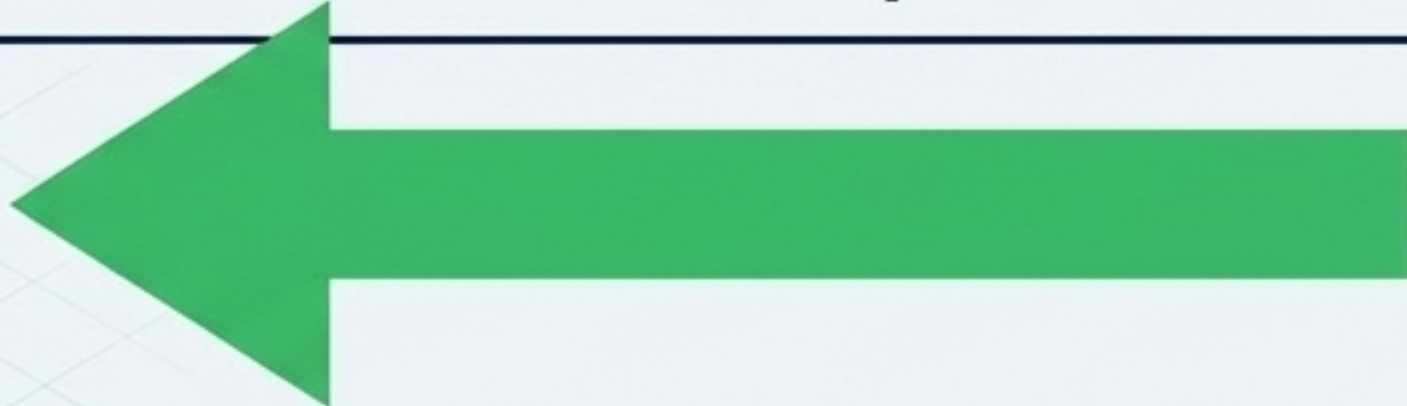


AD-Integrated DNS Architecture & Replication

The Core Concept of AD-Integrated DNS

- DNS data is stored directly within the Active Directory database rather than a separate file.
- Replicates automatically to all Domain Controllers.
- Enables secure dynamic updates directly tied to AD authentication.

The Resolution Paradigm

Forward Lookup Zone	Reverse Lookup Zone
	
Direction: Hostname → IP Address Function: Resolves names (e.g., server1.hp.com) into routable IPs (192.168.1.100). Creation: Automatically generated on Domain Controllers during AD promotion. Primary Record: Host (A)	Direction: IP Address → Hostname Function: Verifies the identity of an IP by translating it back to its system name. Creation: Requires manual initialization. Primary Record: Pointer (PTR)

DNS Zone Archetypes



Primary Zone

Role: The Master Database Copy

Access: Read / Write

Action: Where administrators create, modify, and update resource records.



Secondary Zone

Role: The Backup Copy

Access: Read-Only

Action: Used to distribute DNS load and ensure fault tolerance. Cannot be directly edited.



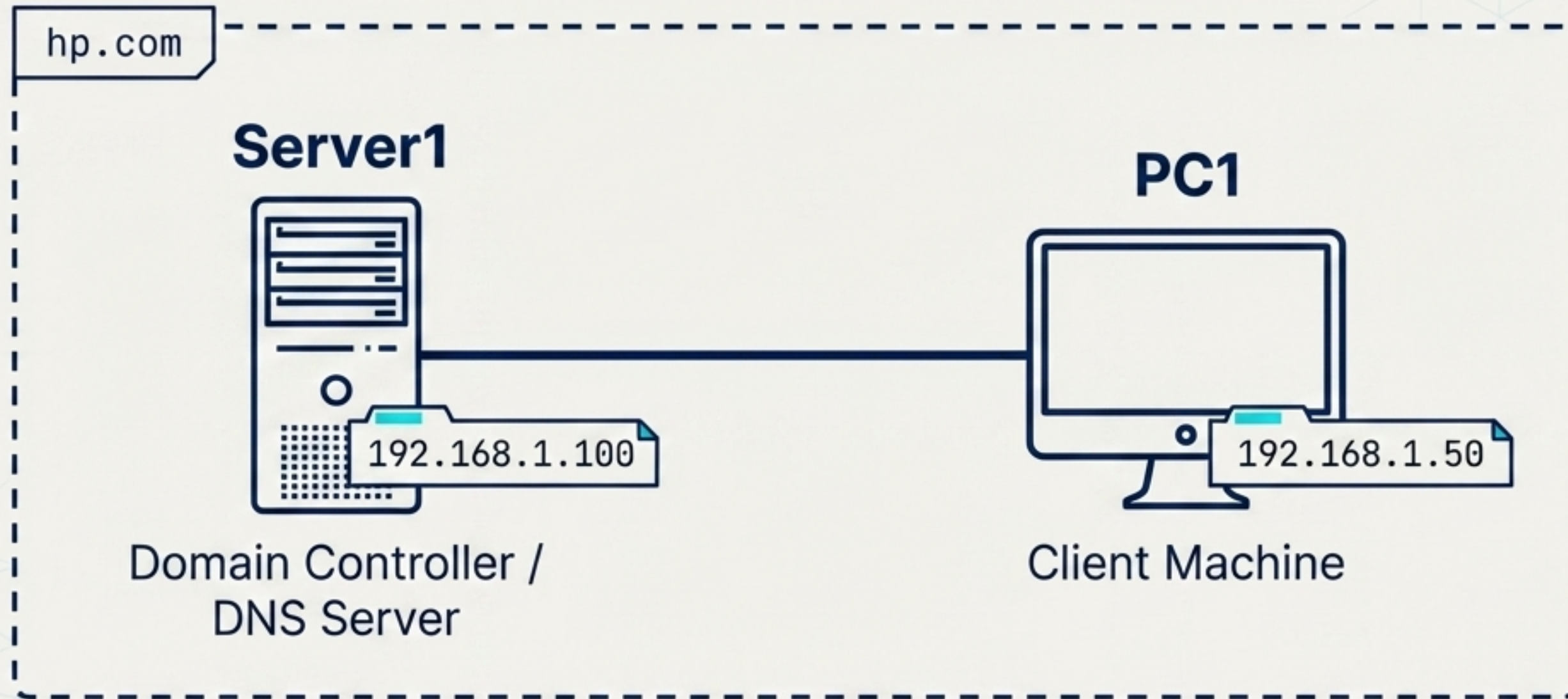
Stub Zone

Role: The Forwarder

Access: Conditional

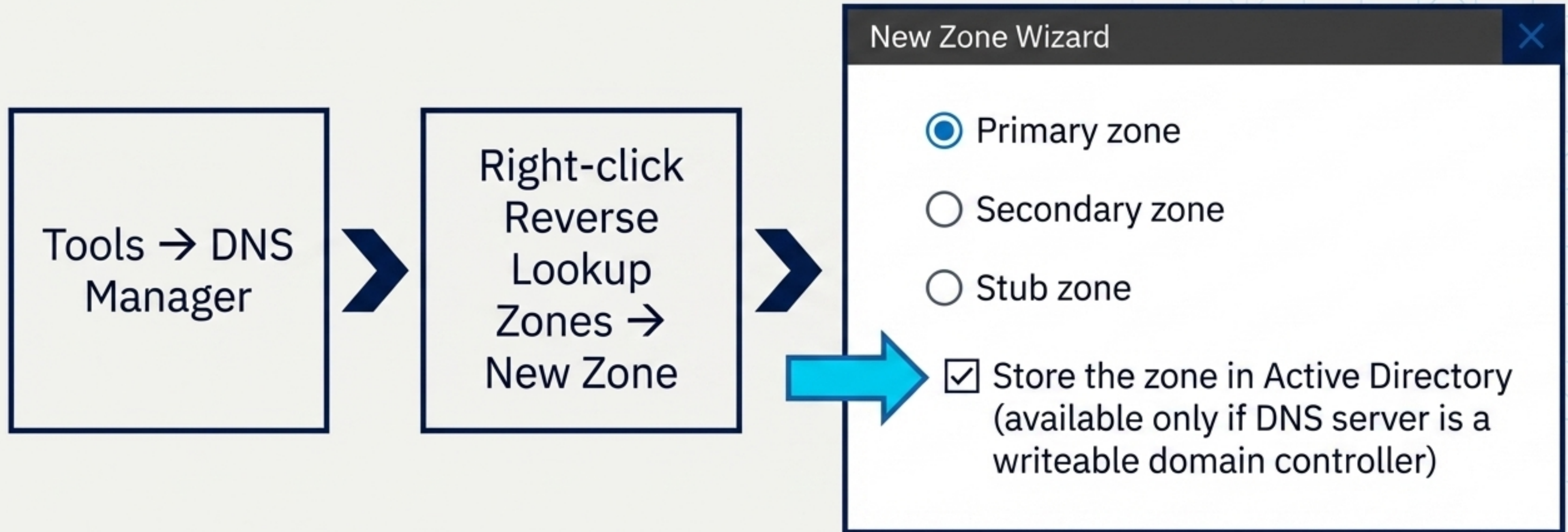
Action: Contains only the records necessary to identify the authoritative DNS servers for a zone.

The Lab Topology



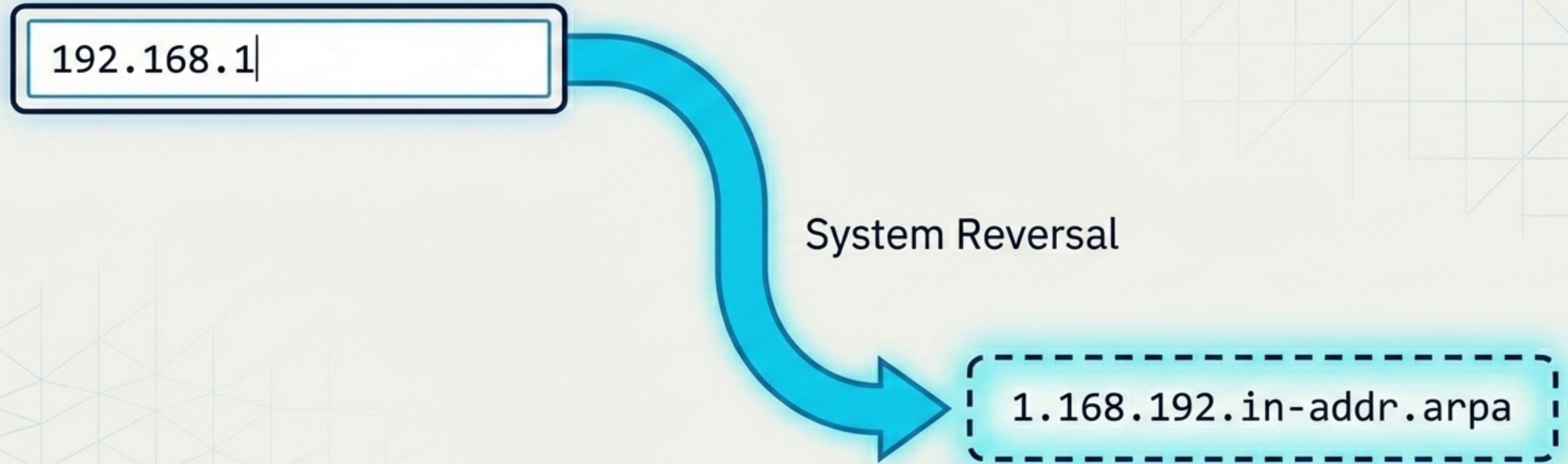
All upcoming configuration steps will establish bidirectional DNS resolution between Server1 and PC1.

Initiating the Reverse Lookup Zone



Selecting 'Store the zone in Active Directory' enables automatic replication across all DNS servers running on DCs in the forest.

Defining the Zone Scope & Network ID



Inputting the IPv4 Network ID automatically generates the specialized `.in-addr.arpa` reverse lookup suffix required by DNS protocols.

Securing Dynamic Updates



A screenshot of a Windows command prompt window showing the 'Dynamic Updates' settings for a DNS zone. The window has a title bar and a menu bar. The 'Dynamic Updates' dropdown menu is open, showing three options: 'None', 'Nonsecure and secure', and 'Allow only secure dynamic updates'. The 'Allow only secure dynamic updates' option is highlighted with a blue background and a white border.

Dynamic Updates
None
Nonsecure and secure
Allow only secure dynamic updates

The Setting:

Restricts DNS updates to authenticated Active Directory clients only.

The Benefit:

Prevents unauthorized devices from hijacking DNS records, ensuring the integrity of the `hp.com` domain routing.

Availability:

This option is only available because the zone is Active Directory-integrated.

Anatomy of DNS Records

A
(Host)

Function: Maps Hostname to IPv4 Address.
Zone: Forward Lookup Zone.

AAAA
(Host)

Function: Maps Hostname to IPv6 Address.
Zone: Forward Lookup Zone.

PTR
(Pointer)

Function: Maps IP Address back to Hostname.
Zone: Reverse Lookup Zone.

Creating Host and Pointer Records

Forward Action: Add A Record

New Host

Name:

PC1

IP address:

192.168.1.50

☒ Create associated pointer (PTR) record 

Reverse Action: Add PTR Record

New Resource Record

Host IP Address:

192.168.1.100

Hostname:

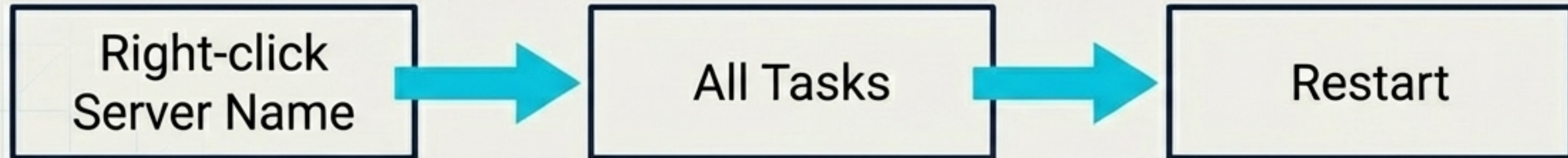
Server1.hp.com

Host records (A) and their reverse mappings (PTR) must be established for both Server1 and PC1 to ensure seamless bidirectional communication.

The Golden Rule of DNS Validation



Always Restart the DNS Service.



Context: Restarting the service forces Active Directory to commit and apply the newly created reverse lookup zones and pointer records before validation testing begins.

Server-Side Validation (nslookup)



```
> nslookup server1.hp.com
Server: localhost
Address: 127.0.0.1
Name: server1.hp.com
Address: 192.168.1.100

> nslookup 192.168.1.50
Server: localhost
Address: 127.0.0.1
Name: pc1.hp.com
Address: 192.168.1.50
```

nslookup queries confirm that both Forward and Reverse Lookup zones are actively translating requests.

Connectivity Validation (ping)

```
> ping pc1.hp.com  
Pinging pc1.hp.com [192.168.1.50] with 32 bytes of data:  
Reply from 192.168.1.50: bytes=32 time<1ms TTL=128  
Reply from 192.168.1.50: bytes=32 time<1ms TTL=128
```



A successful ping to a hostname proves that the system successfully resolved the name to the target IP address (192.168.1.50) prior to sending the ICMP packets.

Client-Side Configuration

For PC1 to utilize the hp.com DNS infrastructure, its primary network adapter must explicitly point back to the Domain Controller's IP address.

Use the following DNS server addresses:
Preferred DNS server: 192.168.1.100

Internet Protocol Version 4 (TCP/IPv4) Properties

General

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

☒ Obtain an IP address automatically

☐ Use the following IP address:

IP address: . . .

Subnet mask: . . .

Default gateway: . . .

☐ Obtain DNS address server automatically

☒ Use the following DNS server addresses:

Preferred DNS server: 192 . 168 . 1 . 100

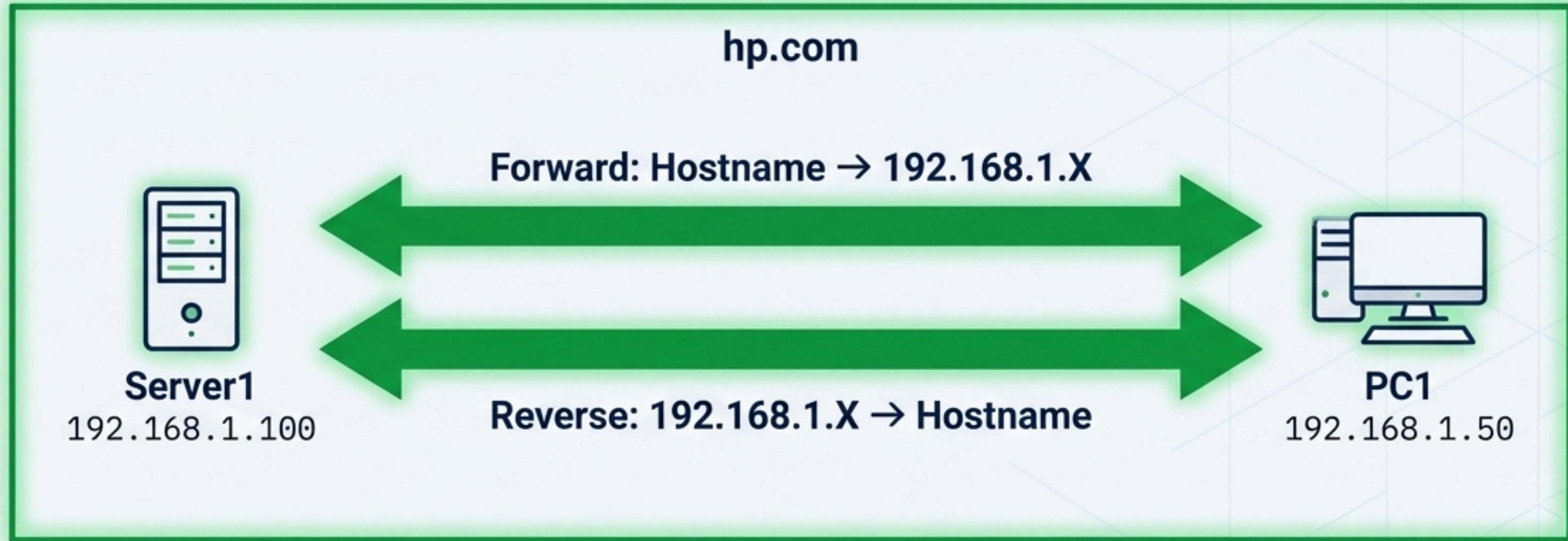
Alternate DNS server: . . .

☐ Validate settings upon exit

Advanced...

OK Cancel

Architecture Complete: Bidirectional Bidirectional Resolution



Active Directory Integrated DNS is fully configured. The network securely translates traffic in both directions, establishing a resilient foundation for the domain.